
Andrés Plaza Aguilar

Faculty of Forest Sciences and Nature Conservation, Universidad de Chile

Av. Santa Rosa 11315, La Pintana, Santiago — Chile

+56 9 7387 1085 | andres.plaza@uchile.cl

Summary

Assistant Professor and researcher at the Faculty of Forest Sciences and Nature Conservation, Universidad de Chile, with experience in data modeling and analysis and research project management. Main research areas involve ecological physiology, sustainable forest production, and monitoring of forest ecosystems. Currently, he is the principal investigator of the Climate Adaptation and Forest Management Laboratory at Universidad de Chile. He has experience leading inter-sectoral research and innovation projects, working in collaboration with private and public institutions such as Empresa Forestal CMPC and the National Forest Service of Chile (CONAF). His teaching activities include data analysis of Forest Ecosystems, technology and production in the forest industry, as well as research methods for undergraduate students. He also teaches technical workshops on data analysis from different sources, such as remote sensing and field measurement data. Dr. Plaza-Aguilar holds a PhD in Plant Sciences from the University of Cambridge, UK, a Master's degree in Functioning and Modelling of Terrestrial Ecosystems from the University of Bordeaux, France, and a degree in Forest Engineering from the University of Talca, Chile. He has extensive experience in fieldwork and laboratory research. He is fluent in Spanish, English, and French.

Education

- **Doctor of Philosophy in Plant Sciences**, University of Cambridge, UK. Physiological Ecology and Forest Conservation Groups. *Thesis: Analysis of leaf phenology of temperate deciduous forest, using digital images from the European phenology network.*
 - **MSc in Ecology, Functioning and Modelling of Terrestrial Ecosystems**, Université de Bordeaux, France. *Master's Thesis: Assessment of cavitation resistance at the intra- and inter-specific level in trees.*
 - **Forest Engineering Degree**, Universidad de Talca, Chile. *Thesis: Analysis of firewood commercialization and its implications for the sustainable use of forest resources, in relation to the ecotourism potential of Navarino Island.*
-

Teaching

- Semester Course in Forest Production (Mandatory)
 - Semester Course in Soil–Vegetation–Atmosphere Interaction (Mandatory)
 - Semester Course in Forest Analysis with R (Elective)
 - Semester Course in Research Workshop and Tutorial Work (Mandatory)
 - Statistics for Natural Resources 2 (Master's level)
-

Work Experience & Collaborations

- **Since January 2017** — Assistant Professor, Department of Forest Management and the Environment, Faculty of Forest Sciences and Nature Conservation, Universidad de Chile.
 - **2022–2024** — Director of the Undergraduate Program in Forest Engineering, Faculty of Forest Sciences and Nature Conservation, Universidad de Chile.
 - **2011–2015** — Research work and collaboration with Forest Research, the research agency of the UK Forestry Commission, specifically with the environmental change research program led by Dr. James Morison; and with the Soil–Plant–Atmosphere Interaction Unit of the National Institute for Agricultural Research (INRA) in Bordeaux, France, specifically with Dr. Lisa Wingate and Dr. Jérôme Ogée.
 - **2010** — Research Assistant at INRA-BIOGECO, Université de Bordeaux, France. Research on genetic variation and phenotypic plasticity in cavitation resistance in trees — Assessment of drought resistance in trees at the inter- and intra-specific level. Project: *Biodiversity And Climate Change, A Risk Analysis* (BACCARA; <https://cordis.europa.eu/project/rcn/89362/factsheet/en>).
 - **2004–2006** — Research Assistant in the BOKONCHIL project on Navarino Island, on the non-market valuation of biodiversity of Navarino Island (Chilean Antarctic Province), funded by the German Federal Ministry of Education and Research (BMBF).
 - **2008–2009** — Engineer in charge of phytosanitary treatment, Empresa Montes Alpha. Certification of timber and packaging materials for export using the ISPM No. 15 standard (16 months).
-

Publications & Conference Presentations

- Martínez, D., Chadwick, C., & Plaza-Aguilar, A. (2023). The Time of Emergence (ToE) of the Andean Mediterranean sclerophyllous forest of *Quillaja saponaria* (Mol.) and *Lithraea caustica* (Mol.) Hox. & Arn. *Forest Ecology and Management*, 544, 121169. <https://doi.org/10.1016/j.foreco.2023.121169>
- Basualto P., Plaza-Aguilar A. (2024). A Comparative Study of Cavitation Vulnerability Among Five *Nothofagus* Species in Chile’s Maule Region. *Annual Meeting of the British Ecological Society*.
- Plaza-Aguilar A., Villalobos M. (2021). Vessel length and cavitation resistance of Sclerophyll woody species from Central Chile: we need more data! *Ecology Across Borders: Annual Meeting of the British Ecological Society and the French Society for Ecology and Evolution (SFE²)*. 10–13 December, Liverpool, UK.
- Plaza-Aguilar A., Ogée J., Filippa G., Cremonese E., Wilkinson M., Morison J., Griffiths H., Wingate L. Comparison between leaf flushing dates of *Fraxinus excelsior* and *Quercus robur* obtained from digital images, with dates predicted from phenology models. (*Submitted*)
- Plaza-Aguilar A., Ogée J., Morison J., Wilkinson M., Mizunuma T., Griffiths H., Grace J., Wingate L. Comparing seasonal differences in the phenology and physiology of oak and ash using canopy colour signals. (*Submitted*)
- Wingate L., Ogée J., Cremonese E. et al. (2015). Observing canopy development and physiology using the EUROPhen camera network at flux sites. *Biogeosciences Discussions*.
- Wingate L., Ogée J., Cremonese E., Filippa G., Mizunuma T., Migliavacca M., Plaza-Aguilar A., Wilkinson M., Grace J. (2015). Opportunities to investigate the functional phenology of ecosystems using a European Phenology Camera Network. EGU 2015, European Geosciences Union General Assembly, Vienna, Austria, 12–17 April 2015.
- Plaza-Aguilar et al. (2014). British Ecological Society and Société Française d’Écologie Annual Meeting, Lille, France. Oral presentation: “Comparing seasonal differences in the phenology of oak and ash trees using canopy colour signals”.

- Saenz-Romero C., Lamy J-B., Loya-Rebollar E., Plaza-Aguilar A., Burlett R., Lobit P., & Delzon S. (2013). Genetic variation of drought-induced cavitation resistance among *Pinus hartwegii* populations from an altitudinal gradient. *Acta Physiologiae Plantarum*.
 - Plaza-Aguilar et al. (2013). *Ecology: Into the Next 100 Years*. London, UK. Poster presentation: “Carbon uptake and water use in a temperate forest: Growing season phenology”.
-

Research Projects

- **Principal Investigator** — “Adaptation to water deficit in xerophytic woody species of central Chile.” — Project funded by the Native Forest Research Fund – CONAF. - Ongoing4
 - **Co-Director** — ANID PINC230012 Project: “Guidelines for wildfire mitigation based on the phenology and seasonal condition of fuels. Applied research for South-Central Chile.” — Ongoing.
 - **Principal Investigator** — “DSS Forest Management: Proposal of a solution suite to support decision-making in forest management for Empresa Forestal Mininco SpA.” — Analysis of plantation establishment, management and weed control (Rango Full, Semazina, among others), fertilization (Multicote, Basacote, among others), site preparation and silvicultural management. — Ongoing.
 - **Co-Investigator** — “Future forests with only pioneer trees? The relevance of species demography to predict forest responses to climate change.” FONDECYT Regular — Ongoing.
 - **Co-Investigator** — “Tree Species Capacity to Recruit in Future Suitable Environments: The Relevance of the Regeneration Niche.” FONDECYT Regular — Completed.
 - **Collaborator** — DOM Seasons: Tracking seasonality in dissolved organic matter. GLEON Network Project — Ongoing.
 - **Principal Investigator** — “Evaluation of the effects of logging trails on the productivity of different Forestal Mininco S.A. sites.” Extension project with Forestal Mininco S.A. — Analysis and productivity monitoring using digital images and satellite products (Sentinel and Landsat). — Completed.
 - **Co-Investigator** — “Alternatives to achieve maximum forest growth potential at sandy sites without water table contribution in Forestal Mininco S.A.” Extension project with Forestal Mininco S.A. — Growth analysis and mortality monitoring using satellite imagery (Sentinel and Landsat). — Completed.
-

Research Skills

- **Programming:** Advanced R Statistics user for GIS; basic knowledge of Python, IDL, and Matlab.
 - **LiCOR 6400** — Measurement of photosynthetic assimilation and stomatal conductance in English Oak leaves.
 - **QGIS, ArcGIS, Google Earth Engine** — Advanced user in Geographic Information Systems.
 - **Hemiview Software** — Estimation of Leaf Area Index from hemispherical photographs.
 - **Sap Flow Systems** — Tissue Heat Balance and Granier method for measuring sap flow in trees.
 - **Spectrophotometry** — Used to calculate pigment concentration in leaves.
-

Languages & Computing

- **Languages:** English and French (advanced level). Basic knowledge of German.
- **Computing:** Regular use of MS Office, ImageJ, and R Statistics.